

3

Level 3 Calculus

91578 Apply differentiation methods in solving problems

PRACTICE ASSESSMENT C (v4)

Achievement	Achievement with Merit	Achievement with Excellence
Apply differentiation methods in solving problems.	Apply differentiation methods, using relational thinking, in solving problems.	Apply differentiation methods, using extended abstract thinking, in solving problems.

You should attempt ALL the questions in this booklet.

Make sure that you have the Formulae and Tables Booklet L3–CALCF.

Show ALL working.

If you need more room for any answer, use the extra space provided at the back of this booklet.

Check that this booklet has pages 2–13 in the correct order and that none of these pages is blank.

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QUESTION ONE

- (a) Differentiate $y = \sqrt{6 - 6x^2}$.

You do not need to simplify your answer.

- (b) Find the gradient of the tangent to the curve $y = \left(2x + \sqrt{x^2 + 1}\right)^2$ at the point where $x = 2$.

You do not need to simplify your answer.

You must use calculus and show any derivatives that you need to find when solving this problem.

(c) Ahead of the City Rail Link's public opening, the Auckland rail network is undergoing safety testing. A train approaches a sharp curve at a speed of 22 m s^{-1} . It takes exactly 15 seconds for the train to pass through this curve.

To navigate the turn safely, the train slows down. The total reduction in speed over time t , in seconds, is modelled by the formula

$$\Delta v(t) = 2 \ln(t + 1) \tan(0.05t), \text{ where } 0 \leq t \leq 15.$$

The maximum safe deceleration for a train on this curve is 0.8 m s^{-2} .

Given that the train experiences its greatest deceleration at the exact moment it leaves the curve, determine whether the train can pass through the curve safely AND find the final speed as it leaves the curve.

Clearly show ALL working.

You must use calculus and show any derivatives that you need to find when solving this problem.



Photo by Johannes Plenio on Unsplash

Source: https://unsplash.com/photos/black-and-brown-chess-piece-on-white-tray_VwkL9LVa-M

(d) Consider the function $f(x) = (kx^3 + 2x^2 + 3)e^{2x}$, where k is a constant. The tangent line to the graph of $f(x)$ at $(-1, 0)$ intersects the y -axis at a height of $\frac{11}{e^2}$.

Find the value of k and then the gradient of the tangent.

You must use calculus and show any derivatives that you need to find when solving this problem.

(e) Consider a curve defined by the pair of parametric equations:

$$x = 27 \cos^3 t \text{ and } y = 27 \sin^3 t, \text{ where } 0 \leq t \leq \frac{\pi}{2}.$$

A tangent line to this curve intersects the x -axis at point A and the y -axis at point B .

Show that the distance between point A and point B is exactly 27.

You must use calculus and show any derivatives that you need to find when solving this problem.

QUESTION TWO

(a) Differentiate $y = \frac{\ln(1 - x^2)}{x^2 + 1}$.

You do not need to simplify your answer.

- (b) A report by He Pou a Rangi (Climate Change Commission) states that urgent and decisive action is needed on how communities will pay for the costs of adapting to climate change.

The amount of funding needed, in thousand dollars, for a particular area, is estimated by the formula $F(t) = 20t(2t + 5)^{\frac{1}{2}}$, where t is the number of months after the present time and $t \geq 0$.

Find the **exact rate** at which the funding needed is increasing after 10 months.

You must use calculus and show any derivatives that you need to find when solving this problem.

- (c) The surface area of a sphere is decreasing at a rate of $12\pi \text{ cm}^2 \text{ min}^{-1}$.

Find the rate of change of the sphere's volume at the instant the radius is 6 cm.

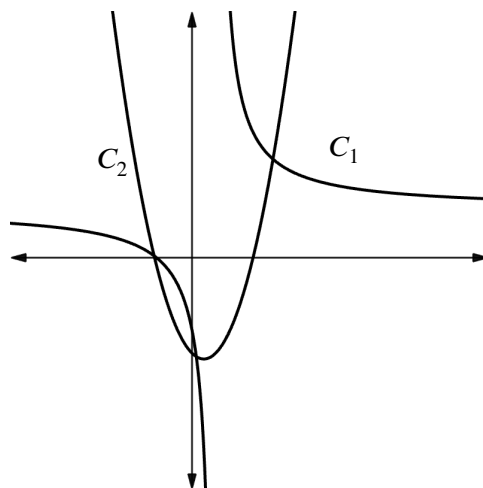
You must use calculus and show any derivatives that you need to find when solving this problem.

*Question Two continues
on the next page.*

- (d) A curve C_1 is defined by the equation $y_1 = \frac{2x+3}{x-1}$, where $x \neq 1$. A straight line L is the tangent to the curve C_1 at the point P , where $x = 2$. The tangent L intersects a second curve C_2 defined by the equation $y_2 = x^2 - x - 4$. The intersection occurs at two distinct points.

Find the exact coordinates of point Q , given that point Q is the point of intersection with a positive x -coordinate.

You must use calculus and show any derivatives that you need to find when solving this problem.

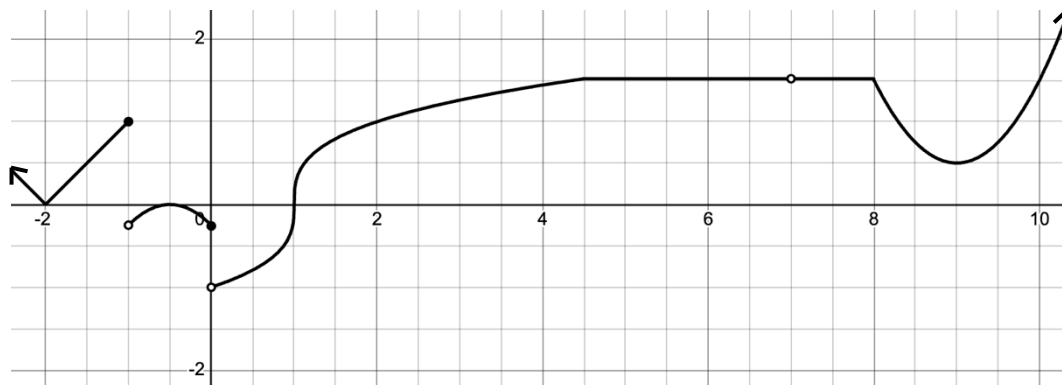


- Show that the rate at which the depth of juice is increasing at the exact moment the juice reaches halfway up the juicer is $\frac{8}{7\pi r^2} \text{ cm s}^{-1}$.



QUESTION THREE

(a) The graph below shows the function $y = f(x)$.



For the function above:

(i) Find the value(s) of x where $f(x)$ is continuous but not differentiable.

(ii) Find the value(s) of x where $f'(x) = 0$.

(iii) Find the value of $\lim_{x \rightarrow -1} f(x)$. State clearly if the value does not exist.

(b) Consider the pair of parametric equations $x = \sec(2t)$ and $y = \cos(t^2)$.

Find an expression for $\frac{dy}{dx}$.

You do not need to simplify your answer.

- (c) Find the x -coordinate(s) of any point(s) of inflection on the curve $f(x) = \frac{x}{\ln(x)}$, where $x > 0$ and $x \neq 0$.

You may assume that any point(s) of inflection found is/are actually a point of inflection.

You must use calculus and show any derivatives that you need to find when solving this problem.

Question Three continues
on the next page.

(d) Two functions are defined as $f(x) = \frac{e^x + e^{-x}}{2}$ and $g(x) = \frac{e^x - e^{-x}}{2}$.

A curve is defined by $y = f(x) \cdot g(x)$, show that $\frac{dy}{dx} = f(2x)$.

You must use calculus and show any derivatives that you need to find when solving this problem.

- (e) An isosceles trapezium has a higher base of length 2, and its two equal non-parallel sides each have length 2, as shown in the diagram.

Find the interior angle θ , between the lower base and one of the equal sides, which will maximise the area of the trapezium.

You **must justify** that the area you have found is indeed a maximum.

You must use calculus and show any derivatives that you need to find when solving this problem.

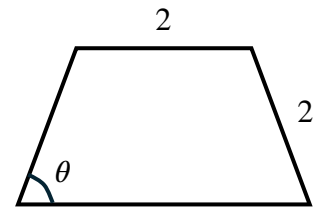


Diagram is NOT to scale